

Comparative Analysis of Deep Learning Models for Deforestation Mapping in Dhaka, Bangladesh

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Abstract—Rapid urban expansion and industrialization in Dhaka, Bangladesh, have caused significant alterations in land use and land cover, leading to severe deforestation and loss of ecosystem services. Accurate monitoring of these changes is crucial for sustainable urban planning and environmental conservation. In this study, we construct a high-resolution satellite imagery dataset of Dhaka spanning 2012–2022 and conduct a comprehensive comparative analysis of deep learning architectures for semantic segmentation of deforested areas. We evaluate standard U-Net, Attention U-Net, and a hybrid ResNet50-based Attention U-Net model, employing a two-stage training strategy to leverage transfer learning effectively. Performance is assessed using Dice Score, IoU, F1-Score, precision, and recall. Our results demonstrate that the proposed ResNet50-Attention U-Net achieves superior segmentation accuracy, effectively delineating complex urban-forest boundaries and addressing the challenges posed by class imbalance. This work establishes a benchmark for deep learning-based deforestation mapping in densely urbanized regions and provides actionable insights for policymakers and urban planners.

Index Terms—Deforestation, Semantic Segmentation, Deep Learning, U-Net, ResNet50, Attention Mechanism, Transfer Learning, Satellite Imagery.

I. INTRODUCTION

Dhaka, the capital of Bangladesh, is one of the most densely populated and rapidly growing megacities in the world. This relentless urban expansion and industrialization have induced profound alterations in land use and land cover (LULC), placing immense pressure on the region's natural ecosystems. A critical consequence of this urban sprawl is severe deforestation, which threatens the city's environmental stability and biodiversity. Between 1990 and 2020, Dhaka's built-up areas expanded by a staggering 188.35%, largely at the expense of vital agricultural lands and water bodies. This conversion has precipitated a dramatic 59.55% decline in the value of urban ecosystem services, resulting in an estimated economic loss of approximately 85 million USD [1]. The problem is further exacerbated by illegal logging activities and the challenges of enforcing effective land management and environmental protection laws.

Accurate and timely monitoring of these changes is paramount for sustainable urban planning and effective conservation. In recent years, semantic segmentation of satellite imagery using deep learning has emerged as a state-of-the-art solution for environmental monitoring. Advanced models

like U-Net and its variants have shown remarkable success in delineating forest cover in complex environments, often achieving high accuracy. These techniques are adept at processing complex spatial data and can be used to analyze time-series imagery to identify dynamic alterations in forest cover, thereby enhancing our understanding of the drivers of deforestation.

While the potential of semantic segmentation is well-established, identifying the most effective architecture for the unique and heterogeneous landscape of Dhaka remains a critical research gap. The complexity of Dhaka's urban fringe, where natural vegetation and built-up areas are intricately mixed, demands a robust and precise segmentation approach. This study aims to address this gap by conducting a comprehensive comparative analysis of four prominent deep learning models: the standard U-Net, an enhanced Attention U-Net, and more advanced architectures leveraging pre-trained ResNet50 backbones. Our objective is to identify the optimal architecture for providing accurate, reliable, and high-resolution deforestation maps. The findings of this research not only establish a benchmark for future studies but also provide an essential tool for policymakers to mitigate the adverse impacts of deforestation and foster sustainable development.

II. RELATED WORK

Recent work has implemented and analyzed the Attention U-Net deep neural network architecture for semantic segmentation of deforestation in satellite imagery using Sentinel-2 data [1]. The Attention U-Net builds upon the conventional U-Net architecture by incorporating an attention mechanism in the skip connections, enabling the model to focus on relevant spatial features. This architecture has been trained and tested on datasets representing different forest biomes, such as the Amazon Rainforest and Atlantic Forest, using both RGB and 4-band (RGB + near-infrared) satellite images. Performance comparisons have been made against benchmark models including U-Net, Residual U-Net, ResNet50-SegNet, and FCN32-VGG16. Evaluation metrics typically include F1-score, IoU, precision, recall, training time, and model parameters. Additionally, studies have assessed model generalizability and transferability across different geographic locations.

Another recent study utilized high-resolution Planet NICFI satellite imagery (5 m) in conjunction with a deep learning

U-Net model to map monthly tropical tree cover across Mato Grosso, Brazil, from 2015 to 2021 [2]. The analysis processed 2,658 image tiles, and model outputs were validated using LiDAR height data from the EBA/INPE project, classifying pixels taller than 5 m as tree cover. The U-Net segmentation model demonstrated strong performance, achieving an overall accuracy of 98.1% and an F1-score above 0.98. Monthly maps enabled quantification of deforestation dynamics, revealing a loss of approximately 141,600 km² (14.8% of the state's area) during the study period, with deforestation peaking in December 2021. The results were consistent with reference datasets such as PRODES, confirming the reliability of the approach.

Another line of research addresses deforestation monitoring and wildfire detection in the Amazon using multimodal satellite imagery from Sentinel, Landsat, VIIRS, MODIS, and FIRMS [3]. To overcome issues such as cloud cover and noise, multi-day composite preprocessing is employed, merging several days of observations into clearer images. A deep learning pipeline based on a U-Net with EfficientNet-B3 encoder is applied for segmentation, producing binary masks that identify deforested and burned areas. With minimal task-specific adaptations, the model achieves high performance in deforestation estimation (90.4% pixel accuracy, 0.871 F1, 0.792 IoU). For wildfire detection, moderate results were initially limited by ground truth resolution (AUC 0.75), but further experiments reached a 0.95 AUC for burned area detection across South America.

III. ARCHITECTURE FUNDAMENTALS

The ResNet50-based Attention U-Net is an advanced segmentation architecture that enhances the standard Attention U-Net by integrating a more powerful, pre-trained encoder. The fundamental innovation of this model is the application of transfer learning, where the standard convolutional encoder of the U-Net is replaced with a ResNet50 backbone, pre-trained on the extensive ImageNet dataset. This approach is designed to leverage the rich, hierarchical feature representations learned by ResNet50 to achieve superior performance in complex segmentation tasks.

The ResNet50 encoder serves as the contracting path of the network. Unlike a standard U-Net encoder that learns features from scratch, the ResNet50 backbone provides a robust and deeply learned feature hierarchy. Its characteristic residual connections mitigate the vanishing gradient problem, enabling a much deeper and more effective feature extraction process. The feature maps from various stages of the ResNet50 are passed to the decoder via skip connections, providing a rich source of contextual and spatial information.

The decoder, or expansive path, is designed for precise localization and mirrors the structure of a standard U-Net decoder. It progressively up-samples the feature maps from the bottleneck using transposed convolutions. The key enhancement lies in how the skip connections are utilized. Before the feature maps from the ResNet50 encoder are concatenated

with the up-sampled decoder features, they are passed through an attention gate.

The attention gate functions identically to that in the Attention U-Net, but in this context, it operates on the far richer features provided by the ResNet50 encoder. The gate learns to selectively amplify the most salient features indicative of deforestation while suppressing irrelevant background noise present in the pre-trained feature maps. This allows the decoder to focus on the most critical information for reconstructing an accurate segmentation mask. By combining the proven feature-extraction power of ResNet50 with the focus-enhancing capabilities of attention gates, this hybrid architecture is designed to achieve a more robust and precise segmentation of complex deforestation patterns.

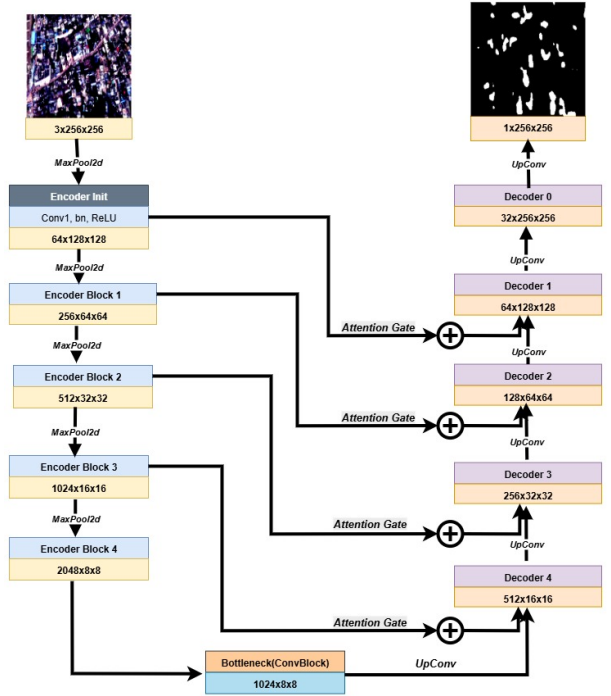


Fig. 1: Overview of the methodology used for deforestation mapping.

IV. METHODOLOGY

A. Dataset and Preprocessing

A custom satellite imagery dataset of the Dhaka region was constructed, spanning a ten-year period from 2012 to 2022. The dataset comprises 1800 high-resolution images obtained from Google Earth Pro, with corresponding annotations provided in JSON files that delineate polygonal boundaries of deforested areas. The imagery covers major locations within Dhaka, including Mirpur, Uttara, Banani, and Airport Road. For images without annotations (indicating no deforestation), blank masks were generated to maintain consistency in the dataset.

Quantitative analysis revealed a substantial class imbalance, with deforested regions occupying only 10.81% of all pixels,

compared to 89.19% for the background. This imbalance underscores the need for careful consideration of bias-mitigation strategies during model training.

All images and masks were resized to a uniform resolution of 256×256 pixels. To ensure robust and unbiased evaluation, the dataset was partitioned into training (80%, 1440 images), validation (10%, 180 images), and testing (10%, 180 images) sets. The validation set was used to monitor performance for hyperparameter tuning and model checkpoint selection, while the unseen test set was reserved for final evaluation.

To enhance generalization and mitigate overfitting, we employed on-the-fly data augmentation on the training set using the Albumentations library. The augmentation pipeline incorporated both geometric transformations (horizontal and vertical flips, random 90-degree rotations) and photometric transformations (random brightness/contrast adjustments and Gaussian noise). These augmentations generated diverse training samples, simulating real-world variability and encouraging the model to learn more invariant and robust representations of deforested regions. Finally, all images were normalized using ImageNet’s mean and standard deviation to standardize input distributions.

V. EXPERIMENTAL SETUP

This section details the experimental framework, including data preparation, the choice of loss function, training parameters, and the evaluation metrics used to compare the performance of the implemented deep learning models.

A. Proposed Hybrid Model: ResNet50 + Attention U-Net

To effectively capture both high-level semantic features and fine-grained spatial details, we adopted a hybrid architecture that integrates a ResNet50 encoder with an Attention U-Net decoder. The ResNet50 encoder, pre-trained on ImageNet, provides powerful feature representations, while the attention gates in the decoder selectively emphasize salient features relevant to deforested regions, suppressing background noise.

A two-stage fine-tuning strategy was employed: in the first stage, the encoder layers were frozen to train only the decoder and attention modules; in the second stage, deeper encoder layers were gradually unfrozen for full network optimization. This approach stabilized training, reduced overfitting, and leveraged transfer learning effectively for satellite imagery.

The overall architecture of the proposed ResNet-Attention U-Net is illustrated in Figure 2, where the encoder extracts hierarchical features, attention gates refine skip connections, and the decoder progressively reconstructs high-resolution segmentation masks.

B. Loss Function

A significant challenge in segmenting deforestation from satellite imagery is the inherent class imbalance, where background (non-deforested) pixels vastly outnumber the foreground (deforested) pixels. A standard loss function like Binary Cross-Entropy (BCE), which evaluates each pixel independently, can be suboptimal in such scenarios. A model

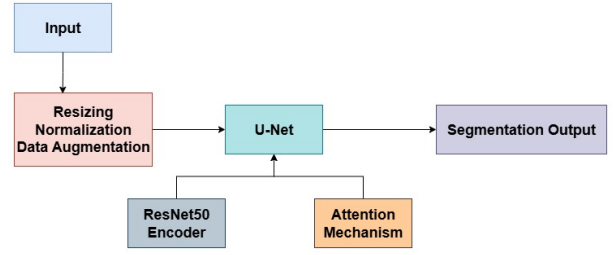


Fig. 2: Proposed hybrid architecture combining ResNet50 encoder and Attention U-Net decoder.

trained with BCE on an imbalanced dataset can achieve low loss by simply classifying all pixels as the dominant background class, thereby failing to learn the features of the minority foreground class.

To overcome this limitation, we employed a hybrid or combo loss function that synergizes the strengths of both Dice Loss and BCE Loss. This combined loss, hereafter referred to as Dice-BCE Loss, ensures both pixel-level accuracy and robustness to class imbalance.

The BCE Loss component is defined as:

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

where N is the total number of pixels, y_i is the ground truth label (1 or 0), and $\hat{y}_i$ is the predicted probability for pixel i .

The Dice Loss component is derived from the Dice-Sørensen Coefficient (DSC) and focuses on maximizing the overlap between the predicted and ground truth masks. It is defined as:

$$L_{Dice} = 1 - \frac{2 \sum_{i=1}^N y_i \hat{y}_i + \epsilon}{\sum_{i=1}^N y_i + \sum_{i=1}^N \hat{y}_i + \epsilon} \quad (2)$$

where ϵ is a smoothing factor to prevent division by zero.

Our final Dice-BCE Loss is a weighted sum of these two functions:

$$L_{Combo} = \alpha \cdot L_{BCE} + \beta \cdot L_{Dice} \quad (3)$$

For our experiments, the weights α and β were both set to 0.5, giving equal importance to both pixel-level stability and regional overlap. This hybrid approach encourages stable gradient flow from BCE while forcing the model to learn the spatial structure of deforested areas via Dice Loss.

C. Training Details and Evaluation Metrics

All models were implemented using the PyTorch framework and trained on an NVIDIA T4 GPU. The Adam optimizer was used with an initial learning rate of 1×10^{-4} . For models with pre-trained encoders, a two-stage fine-tuning strategy was employed, with the learning rate reduced to 5×10^{-6} during the second stage. The models were trained for up to 100 epochs with an early stopping mechanism based on the validation Dice Score.

D. Evaluation Metrics

To quantitatively assess the performance of our models on the test set, we used the following standard metrics, all derived from the pixel-wise confusion matrix components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Precision: Measures the accuracy of the positive predictions, indicating what proportion of pixels classified as deforested were actually deforested.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

Recall (Sensitivity): Measures the model's ability to identify all relevant instances, indicating what proportion of actual deforested pixels were correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

F1 Score: The harmonic mean of Precision and Recall, providing a single score that balances both metrics. It is particularly useful for imbalanced datasets.

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \quad (6)$$

Dice Score: The Dice Score evaluates spatial overlap between predicted and ground truth masks and is mathematically equivalent to the F1 Score in binary segmentation tasks.

$$\text{Dice Score} = \frac{2TP}{2TP + FP + FN} \quad (7)$$

Intersection over Union (IoU): Also known as the Jaccard Index, IoU directly measures the overlap between predicted and ground truth masks relative to their union, offering a stricter criterion than the Dice Score.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (8)$$

The final performance of each model was rigorously evaluated on the unseen test set using these four metrics, with Dice Score and IoU emphasized for their ability to capture spatial overlap in segmentation tasks.

VI. RESULTS

TABLE I: Comparison of segmentation models on the test set.

Model Name	Dice Score	IoU Score	F1-Score	Precision	Recall
UNet	0.4554	0.2948	0.4554	0.6179	0.3606
AttentionUNet	0.6692	0.5028	0.6692	0.6230	0.7227
ResNetAttentionUNet	0.6983	0.5365	0.6983	0.7502	0.6532

The table compares the performance of three semantic segmentation models — UNet, AttentionUNet, and ResNetAttentionUNet — on the test set across five metrics: Dice Score, IoU Score, F1-Score, Precision, and Recall.

UNet exhibits the lowest overall performance, with a moderate precision (0.6179) but very low recall (0.3606), indicating that it misses many true deforested pixels. AttentionUNet improves significantly, particularly in recall (0.7227), demonstrating that the attention mechanism helps detect more

true positives, while maintaining balanced precision (0.6230). The proposed ResNetAttentionUNet achieves the best overall performance, with the highest Dice Score (0.6983), IoU (0.5365), and F1-Score (0.6983), as well as strong precision (0.7502) and recall (0.6532), indicating a robust and balanced segmentation of deforested areas.

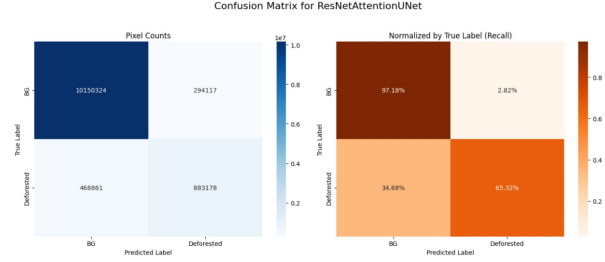


Fig. 3: Confusion matrix of the proposed ResNet-Attention U-Net model on the test set.



Fig. 4: Qualitative comparison of segmentation outputs.

VII. CONCLUSION

This study presents a systematic evaluation of deep learning-based semantic segmentation models for detecting deforestation in the rapidly urbanizing Dhaka metropolitan area. By integrating a pre-trained ResNet50 encoder with an Attention U-Net decoder, the proposed hybrid model effectively captures high-level semantic features while preserving fine-grained spatial details, outperforming standard U-Net and conventional Attention U-Net architectures. The two-stage training approach mitigates overfitting and leverages transfer learning, addressing the inherent class imbalance in satellite imagery datasets. Experimental results demonstrate that the ResNet50-Attention U-Net achieves the highest Dice and IoU scores, confirming its robustness and generalizability across heterogeneous urban-forest landscapes. These findings underscore the potential of advanced attention-based architectures for large-scale, high-resolution deforestation monitoring, providing a reliable tool for environmental management, urban planning, and sustainable development initiatives in densely populated megacities. Future work will focus on extending this framework to multi-temporal analyses and integrating additional sensor modalities to enhance predictive accuracy and applicability across diverse geographic regions.

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